

Kunal Maheshwari

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Summary

Java Backend / SDE2 Engineer with 4+ years delivering high-availability enterprise healthcare and automotive systems using Java, Spring, REST/SOAP, SQL, Azure messaging, and CI/CD.

Delivery scope: Carries changes from requirement clarification and technical design through implementation, API and database integration, testing, code review, documentation, release validation, and production maintenance.

Engineering approach: Designs for reliable message flow, explicit failure behavior, observable service boundaries, and efficient data access; uses performance evidence and authoritative business rules to guide trade-offs.

Working style: Collaborates with product-facing, architecture, engineering, QA, database, deployment, vendor, and operations teams; communicates risks, documents reusable solutions, and mentors developers.

Technical Skills

Java Backend: Java 8/11, Core Java, OOP, Collections, Streams, Spring Framework, Spring MVC, REST/SOAP APIs, Java EE/J2EE, Hibernate/HQL, JDBC/OJDBC; concurrency and multithreading (knowledge); Spring Boot (knowledge).

Azure/Messaging: Azure Service Bus, Azure Relay, Azure Functions, JMS, IBM WebSphere, synchronous REST integration, pub/sub and queue-based workflows; Kafka (project exposure).

Data/DevOps: Oracle, DB2, PostgreSQL (projects/training), SQLite (project); Jenkins, Maven, IBM UrbanCode Deploy, GitLab CI/CD, Git, GitHub Actions; Docker (project), Kubernetes (knowledge).

Testing/AI: JUnit/JUnit4, integration testing, SoapUI, Postman, Log4j2, performance diagnosis, production support, Agile/Scrum; Codex, GitHub Copilot, LangChain, Ollama, RAG, Qdrant, prompt engineering, human-in-the-loop AI validation (projects).

Work Experience

NTT Data, Bengaluru, India

Software Developer, Aug 2022 - Aug 2024

- Developed and maintained **Java 8/11, Spring Framework/JAX-WS, and REST/SOAP** healthcare payer services on IBM WebSphere, supporting **50K+ daily requests at 99.99% SLA availability**.
- Integrated hybrid pub/sub and queue workflows across **Azure Relay, Azure Service Bus, Azure Functions, and JMS**, plus synchronous downstream REST dependencies with explicit timeout and partial-failure behavior.
- Integrated Java services with **Oracle through OJDBC**, consuming established views and PL/SQL procedures while applying member/provider rules and validating downstream data propagation.
- Worked behind an enterprise API Gateway security boundary, consuming authenticated caller identity and enforcing service-level authorization rules.
- **Led a project-wide Log4j2 migration**, owning the PoC, preserving production log contracts, documenting a reusable upgrade path, reviewing changes, and coordinating team rollout.
- Integrated AOP-based service monitoring that propagated correlation IDs and captured **latency, success/failure, and request-volume telemetry** for operational diagnosis.
- Supported root-cause analysis of a **WebSphere memory/resource-retention incident**, contributing code-path review, mitigation strategy, and rolling-restart validation while maintaining service availability.
- Coordinated requirement clarification, implementation, JUnit/SoapUI testing, code review, Jenkins/Maven/IBM UrbanCode workflows, post-release checks, documentation, and go/no-go/rollback recommendations.

Capgemini, Mumbai, India

Senior Analyst, Jul 2020 - Aug 2022

- Developed and supported a Java EE/J2EE, JSF, Hibernate/HQL application on Oracle WebLogic with DB2 for Mercedes/Daimler Vans configuration, recommendation, and ordering workflows.
- Integrated a Python prediction service through asynchronous Java REST requests and callbacks, then enforced authoritative option/package rules before recommendations reached dealer users.
- **Reduced VIN repeat-configuration API latency from 60+ seconds to 10–20 seconds** by moving stable option-code mapping into set-based HQL/DB2 expressions while retaining business validation in Java.
- Diagnosed issues across Python output, Java validation logs, API behavior, and DB2 reference data; used Postman and production logs to isolate defects and verify performance.
- Supported Jenkins/GitLab CI/CD, Git code reviews, JUnit4 testing, Agile delivery, and documentation; **mentored junior developers** through walkthroughs, debugging, and reviews.

Capgemini, Mumbai, India

Analyst Trainee, Jan 2020 - Jun 2020

- Built Java Spring/Spring MVC, Angular, and PostgreSQL training applications with authentication, email/OTP login, CRUD, balances, transactions, and profile workflows.

Projects

CodingHelper - Agentic AI Coding and DSA Platform

- Built a React/TypeScript and Python/FastAPI platform using **LangChain, Ollama/Qwen 2.5 Coder, RAG, Qdrant, and PostgreSQL** to generate and retrieve structured Java/Python solution ladders.
- Implemented isolated Java/Python execution and human-in-the-loop repair, rejection, and promotion gates; generated implementations run against bounded asserting tests before display or reuse.

Article Voice Desk - Local AI Desktop Application

- Built an Electron application with a **shared prioritized inference worker**, streaming document parsers, SQLite persistence, bounded audio/reader windows, locally packaged SmolVLM vision and Supertonic ONNX TTS models, and Windows/macOS release checks.

AWS Automated Spark Pipeline / NYU Taxi Platform

GitHub

- Built an event-driven pipeline where S3 triggers Lambda staging to EFS, a Docker-packaged Spark Structured Streaming workload processes micro-batches on ECS, and data persists to RDS PostgreSQL over JDBC.
- Provisioned AWS networking and services with Terraform; processed an **8GB+ dataset** and persisted checkpoints for restart recovery.

Education

Central Michigan University, Mount Pleasant, MI, USA

Aug 2024 - Dec 2025

Master of Science in Computer Science, GPA: 3.87/4.0

Lovely Professional University, Punjab, India

Aug 2016 - May 2020

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.02/10

Certifications

Microsoft Certified: Azure Fundamentals | Azure Developer Associate | Azure Data Engineer Associate